

# Pipelines Without Pipelines

How we built a data-pipeline engine with **no pipeline object**

*A finite state machine you never wrote, implemented by the scripts themselves*

# First, the thing we're contrasting against

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**Windmill** = run scripts (TS/Python/Go/SQL/...) as jobs.

**Flows** = wire scripts into a workflow.

A flow is a **declarative DAG**, fully defined upfront:

```
FlowValue {  
  modules: [ FlowModule, FlowModule, ... ]  // the whole graph, as data  
}
```

Branches, loops, retries, suspends, sub-flows — all just **data** in that tree.

`backend/windmill-types/src/flows.rs`

# Flows are a finite state machine

Two structures, mirror images:

| Definition (static)              | State (dynamic)                   |
|----------------------------------|-----------------------------------|
| <code>FlowValue.modules[]</code> | <code>FlowStatus.modules[]</code> |
| the graph                        | a cursor + per-step status        |
| <code>flows.rs</code>            | <code>flow_status.rs</code>       |

```
enum FlowStatusModule {  
    WaitingForPriorSteps, WaitingForExecutor,  
    InProgress, Success, Failure, WaitingForEvents,  
}
```

`FlowStatus { step: i32, modules: [...] }` → **persisted as JSONB after every step.**

## The question

The engine is **generic**. The graph lives in the **definition**.

So what if there were **no definition**?

What if the graph were an **emergent property** of the scripts —  
never written down anywhere?

# Data pipelines: there is no pipeline

Flows are **compute-centric** — you orchestrate *runs*. Data pipelines are **asset-centric** — you declare *data*, and the runs follow.

No `PipelineValue`. No DAG file. No orchestrator.

Just scripts that declare what they **touch**:

```
-- producer.sql
ATTACH 'datatable://main' AS db;
CREATE TABLE db.orders AS
  SELECT ... ;

-- consumer.sql
ATTACH 'datatable://main' AS db;
-- on datatable://main/orders
CREATE TABLE db.report AS
  SELECT ... FROM db.orders;
```

`producer` **writes** `orders`. `consumer` **reads** `orders`.

That edge is never declared. It's **parsed out of the SQL**.

# The graph emerges from reads & writes

At deploy, we parse each script and write rows:

```
asset(path, kind, access_type, usage_path)
orders    datatable    W    f/producer
orders    datatable    R    f/consumer
```

The graph endpoint just **SELECT**s and joins these back together.

No compile step. Rename a script → graph updates instantly.

```
[trigger] → producer -writes-▶ (orders) -reads-▶ consumer -▶ (report)
```

# Execution: dispatch, not orchestration

A flow has a conductor stepping `i → i+1`.

A pipeline has **none**. It's a chain reaction:

```
// after ANY top-level script succeeds:
dispatch_asset_triggers(job)
  → for each asset the script WROTE
    → find scripts subscribed to it (// on <asset>)
      → push a job for each
```

`backend/windmill-queue/src/asset_dispatch.rs`

Producer doesn't know its consumers. Consumers don't know each other.

**Subscriptions alone define the graph.**

# How is this even safe to do?

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Both halves need to agree on "what does this script read/write":

- **Backend**, at deploy → must be **authoritative** (it drives real job dispatch)
- **Frontend**, as you type → must be **instant** (live graph in the editor)
- **Testing a draft** → there are no asset rows yet, so the **frontend orchestrates the dev-run** — it's *authoritative* too, before anything is deployed

Two implementations = two sources of truth = drift = your **test** ≠ **prod**.

**So we don't have two implementations.**



# We were already parsing every script

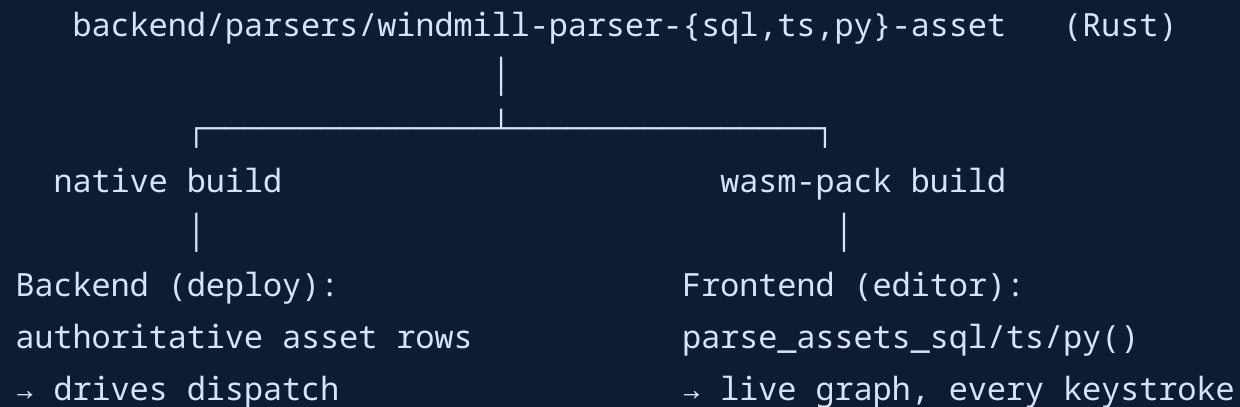
Every Windmill script is **already parsed** to derive its `main` signature — which powers the **input form** (params → JSON Schema → UI) and the **dependencies** (imports → lockfile). That parser already runs **native** (backend) + **WASM** (editor).

```
export async function main(  
  name: string,  
  count = 3,  
  db: Postgresql,           // a Windmill resource type  
) { /* ... */ }
```

↳ a JSON Schema → an auto-form: text field · number (default `3`) · resource picker.

**The eureka:** assets are just *one more visitor* on the same AST — no new infrastructure, just a parser we'd already shipped, hardened, and dual-compiled for years.

# One parser, compiled twice



Same parser, one more visitor. SQL via a real DuckDB-dialect parser; TS/Python via real ASTs (track `ATTACH ... AS db / wmill.datatable('main')`, then resolve usages).

## That's the whole trick

The WASM parser is what makes "no pipeline object" *viable*:

- It can **implement** the graph — backend dispatch off parsed assets
- It can **represent** the graph — frontend draws it live from the same parse

The script definitions *are* the pipeline.

The parser is the bridge that lets one artifact be both **executable** and **drawable**.

## Demo: watch the graph build itself

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1. Open a pipeline folder → **graph reconstructed from a SQL query**
2. Type `CREATE TABLE clean AS SELECT ... FROM orders`  
→ WASM parses → **new node + edge appear, live**
3. Add `// on datatable://main/orders` → **subscription edge wires up**
4. **Deploy** → asset rows written → **run the source**
5. Watch the **cascade animate** down the graph (Activity feed groups the run)
6. Same DAG in the terminal: `wmill pipeline show f/orders`

# The honest tradeoffs

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Emergent graphs are great until they're spooky. So we made the implicit **visible**:

- **Cycle detection** — a write that loops back to its own producer is caught and broken, so cascades can't run away
- **AND-joins**: a script can wait for *all* inputs (partitions via `{partition}` token)
- **dispatch\_event table** logs every decision: *dispatched / join-pending / skipped (why)* — surfaced on the job UI
- Backend re-parses at deploy → **server is authoritative**, browser can't lie
- Self-loops, debounce windows, retries — all declared in-script

# Takeaways

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**Flows** → compute-centric, explicit FSM. A modern **Airflow**.

**Pipelines** → asset-centric, emergent graph. A modern **dbt / Dagster**.

Both on Windmill: **faster, more powerful, polyglot** (TS · Python · Go · SQL · ...).

The unlock: **one parser, native + WASM**, so the script definitions are simultaneously what **runs** and what you **see**.

 **We're hiring** — Rust / TypeScript / systems engineers → [windmill.dev/careers](https://windmill.dev/careers)

Code: `windmill-queue/src/asset_dispatch.rs` · `parsers/windmill-parser-*-asset` · `frontend/.../AssetGraph` — all on `main`.

# Appendix A — The real code

*The actual source behind each claim. Excerpts are verbatim, lightly trimmed where noted.*

## A1 · "Subscriptions alone define the graph"

Straight from the dispatch module's own doc comment — **verbatim**:

```
//! When a script writes an asset and a downstream script subscribes to  
//! that asset via `// on s3://...`, this module pushes a job for each  
//! subscriber after the producer's job completes successfully. Any  
//! asset-writing top-level script cascades – there is no `// pipeline`  
//! gate on the producer side; subscriptions alone define the graph.
```

```
backend/windmill-queue/src/asset_dispatch.rs:11
```



## A2 · The dispatch loop (lightly trimmed)

```
let producers = workspace_producer_writes(db, &job.workspace_id).await?; // cached
let Some(writes) = producers.get(runnable_path).cloned() else { return Ok(..) };

for (asset_kind, asset_path) in writes {
    let trigger_ref = format!("{}", prefix, asset_path);
    let subs = fetch_subscribers(db, &job.workspace_id, &trigger_ref).await?;
    for sub in subs {
        if sub.path == runnable_path { continue; } // skip self-loops
        if sub.join_all { /* AND-join: wait for all inputs */ }
        push_subscriber(db, job, &sub.path, asset_kind, &asset_path,
                        runnable_path, depth + 1, /* partition, debounce, retry */).await?;
    }
}
```

`asset_dispatch.rs::try_dispatch` (273). Non-producers cost **one in-memory lookup, zero queries**.

## A3 · One function, native AND wasm

The **WASM export** is a four-line shim — it forwards to the same crate the backend links natively:

```
[cfg(feature = "asset-parser")]
#[wasm_bindgen]
pub fn parse_assets_sql(code: &str) -> String {
    match windmill_parser_sql_asset::parse_assets(code) { // ← same fn the backend calls
        Ok(r) => serde_json::to_string(&r).unwrap(),
        Err(e) => format!("err: {:?}", e),
    }
}
```

`windmill-parser-wasm/src/lib.rs` — same pattern for `parse_assets_ts` / `parse_assets_py`.

## A4 · ...and it's a real parser

`parse_assets` isn't a regex — it runs the **actual DuckDB-dialect parser** and walks the AST:

```
pub fn parse_assets(input: &str) -> anyhow::Result<ParseAssetsOutput> {
    let statements = Parser::parse_sql(&DuckDbDialect, input)?;
    let mut collector = AssetCollector::new();
    for statement in statements {
        let _ = statement.visit(&mut collector);
    }
    // → reads/writes resolved from ATTACH bindings + table refs
}
```

`backend/parsers/windmill-parser-sql-asset/src/asset_parser.rs`

## A5 · The hard case: data tables in TypeScript

A TS script reaches a data table through the SDK — the parser reads the **TypeScript** *and* the **SQL embedded inside it**:

```
import * as wmill from "windmill-client"
export async function main(since: string) {
  const sql = wmill.datatable("main")      // bind: sql → datatable "main"
  await sql`
    SELECT * FROM orders WHERE created_at > ${since}
  `.fetch()                                // READ   main/orders
  await sql`
    CREATE TABLE daily_revenue AS
    SELECT day, sum(total) FROM orders GROUP BY day
  `.fetch()                                // WRITE  main/daily_revenue
}
```

**Inferred:** `datatable://main/orders` **R** · `datatable://main/daily_revenue` **W** — the TS AST resolves the `sql` binding, lifts each template's SQL, and runs it through the **same SQL parser**. *A parser inside a parser.*

## A6 · The browser side calls it on every keystroke

```
import initAssetParser, {
  parse_assets_ts, parse_assets_py, parse_assets_sql
} from 'windmill-parser-wasm-asset'

export async function inferAssets(language, code) {
  if (language === 'duckdb')      { await initWasmAsset(); return wrap(parse_assets_sql(code)) }
  if (language === 'bun' || 'deno'){ await initWasmAsset(); return wrap(parse_assets_ts(code)) }
  if (language === 'python3')     { await initWasmAsset(); return wrap(parse_assets_py(code)) }
}
```

`frontend/src/lib/infer.ts` — wired into `ScriptEditor.svelte` as a reactive `resource([lang, code])`.

# Appendix B — Drawing the emergent graph

*The graph has no stored layout either — it's computed every render.*

## B1 · Band-reserving tidy-tree layout

The graph is **reconstructed from a SQL query**, so it also has to be **laid out from scratch** each time. Naive layered (Sugiyama) layout let unrelated edges fight for columns. The fix, **verbatim from the source**:

```
// Each node reserves a horizontal band at least as wide as the sum of its
// children's bands ( $W(n) = \max(\text{NODE\_WIDTH}, \sum W(\text{children}) + \text{gaps})$ ) ...
// Bands are exclusive, so two nodes with different parents can never
// interleave horizontally – the failure mode of the previous Sugiyama layout.
//
// The band recursion stops at *join points*: a node with two or more parents
// belongs to no single parent's band ... Joins instead root their own subtree.
```

`frontend/.../AssetGraph/assetGraphLayout.ts` · falls back to a grid on cyclic input.

## B2 · Edges route around obstacles

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A long edge that skips layers would draw **through** unrelated node boxes. So we detour:

- sample the straight line at each intervening row
- if a node sits under it (within `½·width + 8px`), it's a crossing
- route to the side the edge is already heading, outermost lane clears all crossings — **one detour, no box crossings**

`AssetGraphCanvas.svelte::detourForEdge` — computed **once per layout** (`$derived`), not per frame.



## B3 · The live graph is a layered merge

What you see while editing = deployed graph + your unsaved reality, merged by precedence:

```
export function resolveGraph(input) {  
  const acc = seedAccumulator(input, ctx) // 1. base: /assets/graph (deployed)  
  seedDraftOverlays(acc, input) // 2. per-draft seeded triggers & outputs  
  applyLiveBufferOverlay(acc, input, ctx) // 3. the open script's live // on annotations  
  crossCheckSweptScripts(acc, input) // 4. reconcile renamed/removed  
  overlayInferredLineage(acc, input) // 5. WASM-inferred reads/writes (this keystroke)  
}
```

`resolveGraph.ts` — base < session-inferred < draft < **live annotations** (highest wins).